

Exercise 6

In pairs, discuss and answer the following question in your exercise book.

1. List the numbers from 1 to 30.
 - a. Put a circle around each multiple of 3.
 - b. Put a square around each multiple of 4
 - c. List the common multiples of 3 and 4 which are less than 30.
2. Find the lowest common multiple of 2, 3, 4 and 5.
3. Find the least natural number that is exactly divisible by 2, 3, 10, 14 and 15.
4. Three bells ring at intervals of 4, 6 and 9 seconds respectively. If they all ring at 8:00 a.m. at what time will they all ring together again?

Fractions

What is a fraction?

If an orange is cut into four equal parts, each part is called a quarter $\left(\frac{1}{4}\right)$.

A fraction is a part of a whole number. E.g. $\frac{1}{4}$ means one part out of the four parts. The top number is called the **numerator** and the bottom number is called the **denominator**.

$$\text{Fraction} = \frac{\text{Numerator}}{\text{Denominator}}$$

Types of fractions

a. Proper fractions

$\frac{1}{2}, \frac{3}{4}, \frac{7}{10}$ have something in common. What is it?

The numerator is smaller than the denominator. This kind of fraction is called a **proper fraction**.

- b. If the numerator is bigger than the denominator then the fraction is called **an improper fraction**, for example, $\frac{5}{2}, \frac{3}{2}, \frac{10}{7}$.
- c. We can have a whole number and a fraction, for example one and a half oranges. We write this as $1\frac{1}{2}$. This is called a **mixed number** because it consists of a whole number and a fraction.

We can change from improper fractions to mixed numbers and vice versa.

Example

1. Change $1\frac{1}{6}$ into an improper fraction.

$$\text{Solution: } 1\frac{1}{6} = \frac{(1 \times 6) + 1}{6} = \frac{7}{6}$$

2. Change $\frac{5}{3}$ into a mixed number.

$$\text{Solution: } \frac{5}{3} = \frac{3}{3} + \frac{2}{3} = 1 + \frac{2}{3} = 1\frac{2}{3}$$

3. Express $8\frac{3}{4}$ as an improper fraction.

$$\text{Solution: } 8\frac{3}{4} = \frac{(8 \times 4) + 3}{4} = \frac{35}{4}$$

4. Express $\frac{16}{5}$ as a mixed number.

$$\text{Solution: } \frac{16}{5} = 3\frac{1}{5}$$

Comparing fractions

Fractions can only be compared when their denominators are the same.

Example

Arrange $\frac{2}{3}, \frac{4}{5}, \frac{3}{4}$ in ascending order

Find the LCM of 3, 5 and 4

LCM=60

$$\frac{2}{3} = \frac{2 \times 20}{3 \times 20} = \frac{40}{60}, \frac{4}{5} = \frac{4 \times 12}{5 \times 12} = \frac{48}{60}, \frac{3}{4} = \frac{3 \times 15}{4 \times 15} = \frac{45}{60}$$

Arranging the fraction in ascending order (least to greatest), we get

$$\frac{40}{60}, \frac{45}{60}, \frac{48}{60} = \frac{2}{3}, \frac{3}{4}, \frac{4}{5}$$

Exercise 7

1. Write each of the following in numerals:

- Two-thirds
- Three-fifths
- Six-sevenths
- Three-tenths
- Nine-hundredths

2. Find the value of:

- two-thirds of 1 hour
- three-fifths of 1 kg
- one-quarter of 1 ton

3. Copy and insert the missing numbers:

a) $\frac{2}{5} = \frac{6}{\quad}$

c) $\frac{35}{\quad} = \frac{5}{6}$

b) c) $\frac{40}{15} = \frac{\quad}{15} = \frac{2}{3}$

d) $\frac{18}{90} = \frac{\quad}{30} = \frac{3}{\quad} = \frac{\quad}{5}$

4. Write three fractions equivalent to:

a) $\frac{3}{7}$

b) $\frac{2}{5}$

5. Arrange the following fraction in ascending order;

a) $\frac{2}{9}, \frac{4}{5}, \frac{2}{3}$

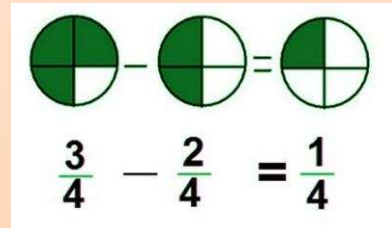
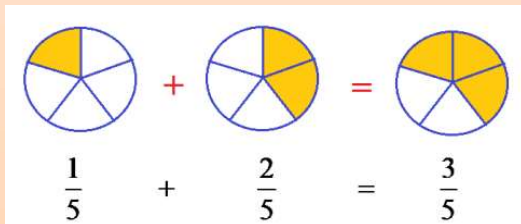
b) $\frac{1}{3}, \frac{2}{5}, \frac{3}{10}, \frac{4}{15}$

Working with fractions

a. Addition and subtraction of fractions

Fractions can be added or subtracted only when their denominators are equal.

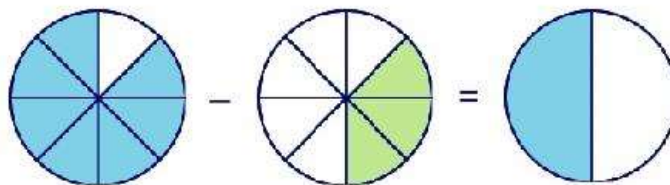
For example, $\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$ and $\frac{3}{5} - \frac{1}{5} = \frac{2}{5}$, only add or subtract the numerator.



$$\frac{7}{8} - \frac{3}{8} = \frac{7-3}{8} = \frac{\cancel{4}^1}{\cancel{8}^2} = \frac{1}{2}$$

Fractions should always be cancelled down to their lowest terms.

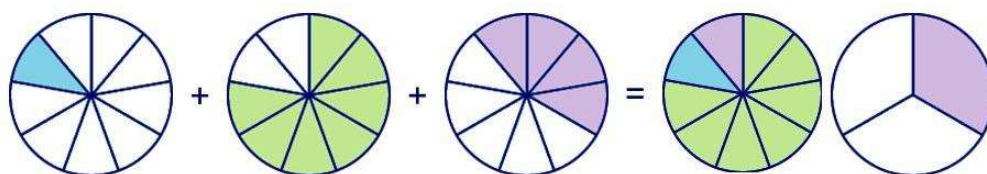
We can show this calculation in a diagram:



$$\frac{1}{9} + \frac{7}{9} + \frac{4}{9} = \frac{1+7+4}{9} = \frac{12}{9} = 1\frac{3}{9} = 1\frac{1}{3}$$

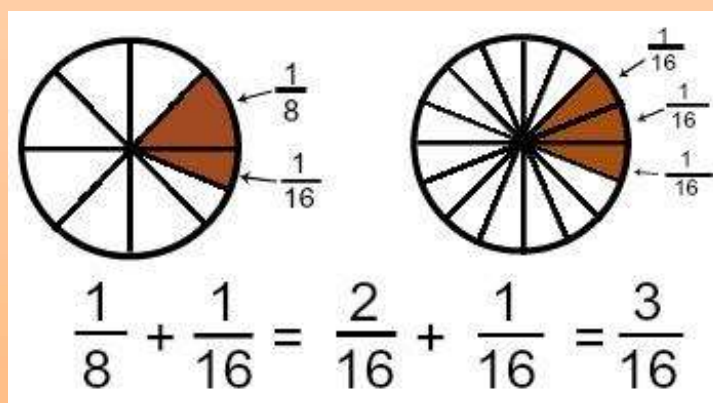
Top-heavy or **improper fractions** should be written as mixed numbers.

Again, we can show this calculation in a diagram:



If the denominators are not the same, then we have to make them the same first by getting the LCM of the numbers, before we add or subtract

$$\frac{1}{3} + \frac{2}{5} = \frac{5+6}{15} = \frac{11}{15} \quad \text{and} \quad \frac{5}{6} - \frac{1}{5} = \frac{25-6}{30} = \frac{19}{30}$$



Activity

In pairs, discuss the following examples and then attempt the questions that follow. The teacher should make sure that the students show understanding by choosing 3 students to explain to the whole class.

Evaluate a) $6\frac{3}{4} + 2\frac{1}{3}$

$$\text{b) } 4\frac{4}{5} - 2\frac{1}{4}$$

$$\text{c) } 4\frac{1}{10} + 2\frac{1}{5} - 2\frac{2}{3}$$

Solutions

$$\text{a) } 6\frac{3}{4} + 2\frac{1}{3} = 8\frac{9+4}{12} = 8\frac{13}{12} = 9\frac{1}{12}$$

- Add the whole numbers
- Find the LCM of 4 and 3 and add the corresponding numerators

- Change the improper fraction $\frac{13}{12}$

into $1\frac{1}{12}$ and add the whole number 1 to 8.

$$\text{b) } 4\frac{4}{5} - 2\frac{1}{4} = 2\frac{16-5}{20} = 2\frac{11}{20}$$

- Subtract the whole numbers
- Find the LCM of the denominators, 5 & 4
- Find the corresponding numerators and subtract.

$$\text{c) } 4\frac{1}{10} + 2\frac{1}{5} - 2\frac{2}{3} = 4\frac{3+6-20}{30} = 4 - \frac{11}{30} = 3\frac{19}{30}$$

- Add & subtract whole numbers: $4+2-2=4$
- Find LCM of 10, 5 & 3 and find corresponding numerators
- Add and subtract.

Exercise 9

Evaluate and simplify:

$$1. \quad 6\frac{1}{2} + 3\frac{3}{4}$$

$$2. \quad 2\frac{7}{8} + 1\frac{4}{5}$$

$$3. \quad 4\frac{1}{5} + 2\frac{3}{10}$$

$$4. \quad 5\frac{3}{4} - 2\frac{7}{8}$$

$$5. \quad 3\frac{1}{2} - 2\frac{3}{4}$$

$$6. \quad 10\frac{7}{16} - 3\frac{3}{8}$$

$$7. \quad -\frac{2}{3} - \frac{4}{5} + \frac{3}{10}$$

$$8. \quad 3\frac{1}{6} - 2\frac{1}{3} + \frac{7}{12}$$

$$9. \quad -2\frac{6}{13} + 1\frac{1}{2}$$

$$10. \quad 4\frac{1}{6} + 1\frac{11}{20} - 3\frac{7}{15}$$

$$11. \quad 10\frac{1}{2} - 3\frac{2}{3} + 4\frac{1}{5} + 2\frac{3}{4}$$

Multiplication of fractions

To multiply two fractions, we multiply the two numerators to get the new numerator and multiply the two denominators to get the new denominator.

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a'c}{b'd}$$

Example

$$a) \quad \frac{3}{5} \cdot \frac{8}{7}$$

$$b) \quad \frac{2}{3} \cdot \frac{6}{11}$$

$$c) \quad 2\frac{7}{10} \cdot 1\frac{2}{3}$$

Solutions

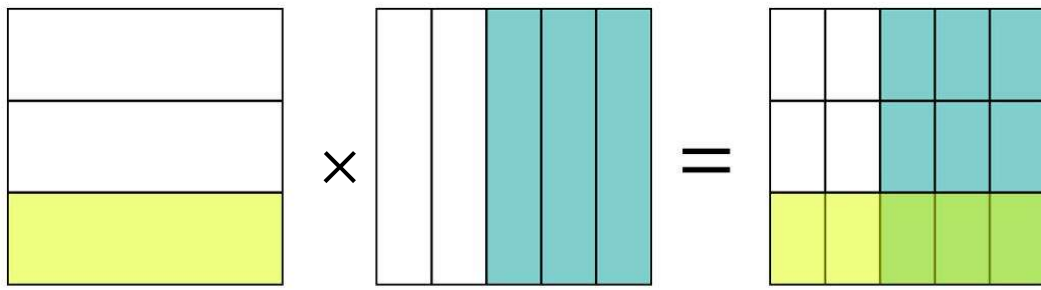
$$a) \quad \frac{3}{5} \cdot \frac{8}{7} = \frac{24}{35}$$

$$b) \quad \frac{2}{3} \cdot \frac{6}{11} = \frac{12}{33} = \frac{4}{11}$$

$$c) \quad 2\frac{7}{10} \cdot 1\frac{2}{3} = \frac{27}{10} \cdot \frac{5}{3} = \frac{9}{2} = 4\frac{1}{2}$$

To help make multiplication easier, we can cancel any common factors in the numerator and denominator before multiplying: $\frac{2}{3} \times \frac{6}{11} = \frac{2}{1} \times \frac{2}{11} = \frac{4}{11}$

Mixed numbers should be first changed to improper fractions before multiplying.



$$\frac{1}{3} \times \frac{3}{5} = \frac{3}{15} = \frac{1}{5}$$

Exercise 10

Working in groups evaluate the following showing all your working clearly and simplifying where possible

1. Find:

a. $\frac{2}{5} \times \frac{3}{4}$

c. $\frac{4}{27} \times \frac{21}{16}$

b. $-\frac{5}{4} \times \frac{2}{3}$

d. $1\frac{1}{2} \times \frac{1}{3}$

2. Find:

a. $\frac{2}{3} \times \frac{1}{4} \times \frac{3}{5}$

d. $\frac{2}{3} + \frac{3}{4} \times \frac{2}{3}$

b. $\frac{3}{8} \times \frac{2}{3} \div \frac{3}{5}$

e. $\frac{3}{5} - \frac{1}{3} \times \frac{6}{7}$

c. $\frac{2}{5} \times 4 \times 1\frac{5}{8}$

f. $\frac{2}{3} + \frac{3}{4} \times \frac{2}{3} - \frac{1}{2}$

Reciprocals

Task

See if you can solve these problems in your head by multiplying the first number by the reciprocal of the second number:

1. $8 \div \frac{1}{5}$
2. $10 \div \frac{1}{10}$
3. $3 \div \frac{3}{8}$

(Hint: All the answers will be whole numbers.)

Clearly, $3 \times \frac{1}{3} = 1$

$\frac{1}{3}$ is called the reciprocal of 3.

3 is called the reciprocal of $\frac{1}{3}$.

One number is the **reciprocal** of another if their product is 1.

The reciprocal of a fraction is obtained by interchanging the numerator and the denominator, i.e. by inverting the fraction.

$$\frac{a}{b} \times \frac{b}{a} = 1$$

Therefore $\frac{a}{b}$ is the reciprocal of $\frac{b}{a}$

For example, the reciprocal of $\frac{5}{6}$ is $\frac{6}{5}$ since $\frac{5}{6} \times \frac{6}{5} = 1$.

c. Dividing fractions

$\frac{1}{2} \div \frac{1}{8} = 4$

Picture: $\frac{1}{2} \div \frac{1}{8} = 4$ 	Another way: $\frac{1}{2} \div \frac{1}{8} = 4 \rightarrow 4 \times \frac{1}{8} = \frac{1}{2}$
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$$\frac{1}{2} \div \frac{1}{8} = \frac{\frac{1}{2}}{\frac{1}{8}} = \frac{\frac{1}{2} \times 8}{\frac{1}{8} \times 8} = 4$$

To divide fractions take the reciprocal (invert the fraction) of the divisor and multiply the dividend.

Invert the fraction that you are dividing by	$\frac{4}{5} \div \frac{2}{3} = \frac{4}{5} \times \frac{3}{2}$
Multiply the numerators and denominators	$\frac{4}{5} \times \frac{3}{2} = \frac{12}{10}$
Simplify the fraction if necessary	$\frac{12}{10} = 1\frac{1}{5}$

This is the quickest technique for dividing fractions. The top and bottom are being multiplied by the same number and, since that number is the reciprocal of the bottom part, the bottom becomes one.

How to Make a Fraction on a Scientific Calculator

By default, scientific calculators, like regular ones, display fractions as decimals. So if you enter a simple fraction, such as $\frac{1}{2}$, the display reads 0.5. Some – but not all – scientific calculators offer a feature that allows you to display fractions without making the conversion. Using this feature, you can enter a complex fraction and simplify it right on your calculator. Calculators with this feature also allow you to enter a number composed of an integer and a fraction, such as $1\frac{1}{4}$. If your calculator doesn't have this feature, you can use a workaround to manipulate fractions.

Exercise 11

In pairs, discuss how you would divide the following fractions. Use a calculator.

a. $\frac{2}{3}, \frac{1}{6}$

b. $1\frac{1}{3}, 3\frac{1}{2}$

c. $2\frac{3}{4}, \frac{2}{3}$

d. $\frac{4}{5}, 3$

e.
$$\begin{array}{r} 2 + \frac{1}{3} \\ 2 - \frac{1}{3} \end{array}$$

f.
$$\begin{array}{r} \frac{2}{3} + \frac{3}{4} \\ \frac{2}{3} - \frac{3}{4} \end{array}$$

g.
$$\begin{array}{r} \frac{1}{3} + \frac{1}{4} \\ 1 - \frac{1}{5} \end{array}$$

Fractions, percentages and decimals

Changing fractions to decimals

You can convert all fractions to decimals. The decimal forms of rational numbers either end or repeat a pattern. To convert fractions to decimals you just divide the top by the bottom — divide the numerator by the denominator — and if the division is not exact, you can stop after a certain number of decimal places and either round or truncate.

Some fractions can be expressed as exact decimals.

$$\frac{1}{2} = \frac{5}{10} = 0.5$$

$$\frac{1}{4} = \frac{25}{100} = 0.25$$

$$\frac{3}{5} = \frac{6}{10} = 0.6$$

$$\frac{7}{10} = 0.7$$

Recurring decimals

Some fractions cannot be expressed as exact decimals.

$$\frac{2}{3} = 0.6666\dots (6 \text{ is repeated}) = 0.66 \text{ to 2d.p. (truncated)} = 0.67 \text{ to 2d.p. (rounded)}$$

$$\frac{5}{6} = 0.8333\dots (3 \text{ is repeated}) = 0.83 \text{ to 2d.p. (truncated)} = 0.83 \text{ to 2d.p. (rounded)}$$

$$\frac{1}{7} = 0.142857142857\dots (142857 \text{ is repeated}) = 0.143 \text{ to 3d.p. (rounded)}$$

Decimals in which one or more figures are continually repeated in the same order are called **recurring or repeating decimals**. They are written by placing a dot over the recurring figure, or dots on the first and last digits of the repeating block of digits.

$$\frac{2}{3} = 0.\dot{6}$$

$$\frac{5}{6} = 0.8\dot{3}$$

$$\frac{1}{7} = 0.\dot{1}4285\dot{7}$$

Changing recurring decimals into fractions:

Example

a) $0.\dot{6}$

b) $0.8\dot{3}$

Solution

a) Let $r = 0.6666\dots$
 $10r = 6.6666\dots$

- ☐ Multiply by 10 because the recurring number is in the **tenths** position.
- ☐ Subtract r from $10r$ to get rid of the decimal part of the number
- ☐ We are then able to find r as a fraction

$$10r - r = 9r$$

$$6.6666... - 0.6666... = 6$$

$$\text{Therefore, } 9r = 6 \text{ hence } r = \frac{6}{9} = \frac{2}{3}$$

b) Let $r = 0.833333...$

$$10r = 8.33333...$$

$$100r = 83.33333...$$

$$100r - 10r = 90r$$

$$83.3333... - 8.3333... = 75$$

$$90r = 75 \text{ hence } r = \frac{75}{90} = \frac{5}{6}$$

- Multiply by 100 because the recurring number is in the **hundredths** position
- To get rid of the decimal part of the number, subtract $10r$ from $100r$
- We are then able to find r as a fraction

Changing decimals to fractions

All terminating and recurring decimals can be expressed as fractions.

$$0.5 = \frac{5}{10} = \frac{1}{2}$$

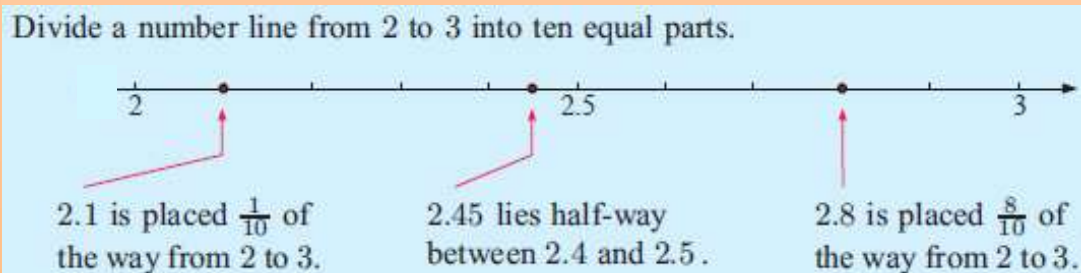
$$0.32 = \frac{32}{100} = \frac{8}{25}$$

Decimal places

Ordering Decimal Numbers

Just like whole numbers, decimal numbers can be shown on a number line.

For example: 2.1, 2.45 and 2.8 can be placed on number line:



2.1 is the least while 2.8 is the greatest.

A number with lots of decimal digits may be less than a number with fewer decimal digits e.g. $1.4 > 1.32568$

Rounding

What is 1.32568 to 3 decimal places?

The numbers **with 3 decimal places** closest to 1.32568 are **1.325 and 1.326**

As 1.32568 is closer to 1.326, $1.32568 = 1.326$ to 3d.p.

What is 1.5 to the nearest whole number?

1.5 is exactly half way between 1 and 2 the convention is to round up to 2.

Exercise 12

1. In a closing-down sale, a shop offers 50% off the original prices. What fraction is taken off the prices?
2. In a survey, one in five people said they preferred a particular brand of milk. What is this figure as a percentage?
3. Keji is working out a problem involving $\frac{1}{4}$. She needs to enter this into a calculator. How would she enter $\frac{1}{4}$ as a decimal on the calculator?
4. Mr. Achan pays tax at the rate of 25% on his income. What fraction is Mr. Achan's income in this?
5. When Deng was buying his plot of land he had to put down a deposit of $\frac{1}{10}$ of the value of the piece of land. What percentage was this?
6. I bought a coat in the December sales with $\frac{1}{3}$ off the original price. What percentage was taken off the price of the coat?
7. Mary bought some fabric that was 1.75 metres long. How could this be written as a fraction?